OUTREACH AGENT

Scoring Rubric

Prepared by MS&E108 Group

1. Purpose and Scope

This rubric provides an objective framework for evaluating emails and LinkedIn messages generated by the NEA Outreach Agent. It serves two evaluation modes:

- **Human evaluation:** Investors and QA reviewers use this rubric to grade agent outputs on a consistent scale
- **LLM-as-a-Judge:** The rubric criteria are designed to be machine-parseable so that a grading LLM (e.g., Claude Opus or GPT-4) can score outputs programmatically using the same standards. This will be our primary scoring methodology.

Design principles: Criteria are self-contained, as each item includes all information needed to evaluate it and anchored to observable features of the output instead of subjective impressions. This follows best practices from Mercor's rubric design framework and OpenAI's HealthBench methodology, which we took inspiration from.

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2. Evaluation Context Tags

Before scoring any output, the evaluator must record the following metadata tags. These normalize scores across different generation conditions and prevent unfair comparisons

Tag	Values	Why It Matters
Data Richness	Rich / Moderate / Sparse	Sparse-data companies hit the cold_with_firm_context fallback, and penalizing weak personalization is unfair when data is unavailable.
Feedback State	Baseline (0 examples) / Early (1–10) / Mature (11+)	Promoted few-shot examples improve voice fidelity over time. Day-0 outputs should be scored more leniently than Day-60 ones
Investor Context	None / Partial / Full	Investor-provided outreach_goal, event details, or prior relationship context improve output relevance
Investor	Ashley / Tiffany / Danielle / Madison	Voice fidelity is investor-specific. A correct output for Madison is wrong for Danielle.
Format	Email / LinkedIn	LinkedIn has a ~300-character practical limit that is not system-enforced

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3. Global Scoring Scale

Each dimension in Section 4 is scored on a 1–5 integer scale. The global score is a weighted average across all dimensions. We round to one decimal place.

Score	Label	Definition
5	**Excellent**	Investor would send this message unedited. No detectable AI tells, strong personalization, perfect voice match.
4	**Good**	Minor adjustments needed (e.g., slightly wrong length, one weak signal choice). Investor would edit 1–2 sentences before sending.
3	**Acceptable**	Structurally correct and on-voice, but personalization is shallow or partially generic. Investor would need to rewrite a paragraph before sending.
2	**Below Standard**	Noticeable problems: wrong voice register, AI tells present, weak structure. Investor would not send this without major edits.
1	**Poor**	Fundamental failures: wrong investor voice, fabricated claims, wrong contact, or structurally incoherent. Requires complete rewrite.

4. Scoring Dimensions

The rubric evaluates eight dimensions. Each dimension has a weight reflecting its importance to the core success metric: “would the investor send this unedited?”

#	Dimension	Weight	Rationale
D1	**Voice Fidelity**	10%	A message that doesn’t sound like the investor is generic
D2	**Personalization Depth**	15%	The primary differentiator between a mass email and a compelling cold outreach
D3	**Context Type Appropriateness**	10%	Upstream detection error; evaluators should flag but weight lower since it’s a pipeline issue, not a generation issue.
D4	**AI-Tell Absence**	25%	Detectable AI patterns destroy credibility with sophisticated founders.
D5	**Factual Accuracy**	30%	
D6	**Technical Quality**	5%	Primarily applies to Ashley and Madison. Weight is conditional: 0% for Danielle consumer emails.
D7	**Actionability**	5%	The email’s purpose is to get a reply. A clear, correctly-styled ask is the final requirement.

Dimension 1: Voice Fidelity

Does the message sound like it was written by this specific investor, not a generic VC or a different investor on the team?

ID	Criterion	Scoring Guide
V1	Greeting style matches investor profile	5: Madison uses em-dash greeting ("James—"); Danielle uses warm "Hey [Name],"; Tiffany/Ashley use "Hi [Name]," 3: Correct formality level but not the investor's specific pattern 1: Wrong style entirely (e.g., "Hi James," from Madison, or em-dash from Danielle)
V2	Message length is appropriate for investor + context type	5: Madison 4–6 sentences; Ashley length proportional to data richness; Tiffany medium; Danielle medium-long with firm block 3: Within ±30% of expected length 1: Gross mismatch (Madison at 300+ words; Ashley at 3 sentences on a thesis topic)
V3	Formality register matches investor voice	5: Danielle casual/warm; Tiffany professional/efficient; Ashley intellectually warm; Madison concise/credential-forward 3: Register is close but includes 1–2 off-register phrases 1: Register is wrong (e.g., corporate tone for Danielle, casual for Tiffany)
V4	Ask style matches investor pattern	5: Ashley = soft collaborative; Tiffany = concrete + NYC option; Danielle = inclusive team; Madison = one direct sentence 3: Ask is present and reasonable but uses generic phrasing rather than the investor's pattern 1: Wrong ask style entirely or missing ask
V5	Sign-off matches investor conventions	5: Correct sign-off name and formality (e.g., "Best, Ashley" or just "Ashley") 3: Name is correct but formality slightly off 1: Wrong name, missing sign-off, or grossly wrong formality
V6	Firm context block usage is correct	5: Danielle appends full firm context block; Madison includes when data sparse; Ashley/Tiffany generally omit 3: Block present/absent with minor deviation from expected pattern 1: Block included when it shouldn't be, or missing when required (Danielle)

Dimension 2: Personalization Depth

How specific is the personalization to this founder and company? Scored against the system's 7-tier priority hierarchy.

Personalization Priority Hierarchy (Reference)

Tier	Signal Type	Example
P1	Specific technical detail	Architecture choice, named feature, design decision
P2	Published paper or blog reference	Shows homework beyond the company page
P3	Shared background	Same employer, university, research area (must be verifiable)
P4	Recent milestone	Funding, launch, hire, partnership (timely and concrete)
P5	Market thesis alignment	Investor has active thesis that fits; connection explained
P6	Portfolio relevance	Complementary portfolio company with meaningful connection

ID	Criterion	Scoring Guide
P1	Highest-tier signal is used from available data	5: Uses highest available tier (e.g., P1 technical detail when data supports it) 3: Uses a mid-tier signal when a higher-tier was available in the data 1: Uses only P6–P7 signals when P1–P3 data was available, or no personalization at all
P2	Signal is specific to THIS founder/company	5: Reference would only make sense for this specific company (e.g., a named product feature) 3: Reference is sector-appropriate but could apply to 5+ companies 1: Reference is fully generic ("impressive product", "love what you're building")
P3	Every personalization claim is traceable to provided data	5: Each signal can be mapped to a specific field in the CompanyBundle 3: Most signals are traceable; one is a reasonable inference 1: One or more signals appear fabricated or unsupported by data
P4	Portfolio mentions include a meaningful connection	5: Portfolio company is named with a clear, specific reason it's relevant 3: Portfolio company named with a vague connection ("we've invested in this space") 1: Portfolio dropped without any connection to the founder's company (explicit anti-pattern)

Dimension 3: Context Type Appropriateness

Was the correct context type selected for the available data and investor intent? This dimension evaluates an upstream pipeline decision, not the LLM's writing quality.

ID	Criterion	Scoring Guide
C1	Context type matches the strongest available signal	5: Best-fit type selected (e.g., cold_funding_congrats fires when funding_announcement is the top signal) 3: Reasonable type but not optimal (e.g., cold_technical_interest when thesis_driven_deep_dive was better) 1: Clearly wrong type (e.g., cold_personal_affinity for a deep-tech infrastructure company)
C2	Warm/event types fire when investor context is provided	5: has_prior_relationship=True correctly triggers warm type; has_event_context=True correctly triggers event type 3: N/A (binary: either the flag was set and it fired, or it wasn't) 1: Flag was set but warm/event type did not fire (pipeline bug)
C3	Fallback (cold_with_firm_context) did not fire when richer signals were available	5: Non-fallback type selected with clear justification from signals 3: Fallback fired but signals were only marginally above threshold 1: Fallback fired despite multiple strong signals being present
C4	Opening move matches context type	5: Opens correctly (e.g., hook for thesis_driven_deep_dive, congrats for cold_funding_congrats, credential for cold_market_gap) 3: Opens with a relevant but non-canonical move 1: Opens with the wrong type entirely (credential when it should be a hook, or generic when it should be congrats)
C5	Body sequencing follows the email pattern	5: All body elements appear in the correct order per the context type's email_pattern 3: Elements present but in slightly wrong order 1: Key structural elements missing or misordered
C6	Self-introduction is correctly positioned	5: Self-intro comes second (not first) for Ashley; comes first for Tiffany; credential-forward for Madison 3: Self-intro present but positioned one step off 1: Self-intro missing or in a position that breaks the email's logic
	Subject line present and appropriate (email format only)	5: Subject line present, specific, and non-generic 3: Subject line present but somewhat generic

		1: Subject line missing (parsing failure) or misleading
C7	LinkedIn format respects practical length limits	5: LinkedIn message is concise, under ~300 characters for preview readability 3: Slightly over preview limit but still readable 1: LinkedIn message is email-length with no format adaptation (Score N/A if format is email)

Dimension 4: AI-Tell Absence

Each detected AI tell deducts from a starting score of 5.

Scoring method: Start at 5. Deduct 1 point for each major violation and 0.5 for each minor violation. Floor at 1.

ID	AI Tell	Severity	Example
A 1	Em-dash used mid-sentence (not in Madison greeting)	**Major (−1)**	*"The architecture — combining retrieval with traversal — is compelling"*
A 2	Generic opener from banned list	**Major (−1)**	*"I came across your company and was impressed"*
A 3	Prior employment reference	**Major (−1)**	*"As a former Meta researcher, you understand..."*
A 4	Vague compliment with no specifics	**Major (−1)**	*"I love the vision here"*
A 5	Salesy or hype language	**Major (−1)**	*"exciting opportunity", "game-changing", "revolutionary"*
A 6	Fabricated or unsupported specifics	**Major (−1)**	*Technical claim not in the CompanyBundle data*
A 7	Exclamation inflation (2+ more than investor norm)	**Minor (−0.5)**	*5 exclamation marks when investor style uses 0–1*
A 8	Over-formal opener for warm investor	**Minor (−0.5)**	*"I hope this message finds you well" from Danielle*
A 9	Copied sentence from few-shot example	**Minor (−0.5)**	*Exact phrasing recycled from email_samples.md*
A 10	Wrong greeting style for investor	**Minor (−0.5)**	*"Hi James," from Madison (should be "James—")*

Dimension 5: Factual Accuracy

Are all claims in the email traceable to the actual data provided in the CompanyBundle? This uses a violation-counting approach similar to D5.

Scoring method: Start at 5. Deduct 2 points for each major factual error and 1 point for each minor error. Floor at 1. A single factual error triggers an automatic flag.

ID	Criterion	Scoring Guide
F1	No fabricated technical details or product claims	Major (−2): Technical architecture, feature, or capability mentioned that is not in the data Minor (−1): Slightly embellished description that is a reasonable inference but not directly stated
F2	No incorrect portfolio claims	Major (−2): Portfolio company attributed to NEA that is not in the investor's portfolio list Minor (−1): Correct portfolio company but wrong context (e.g., wrong round or wrong relationship)
F3	No wrong funding round details	Major (−2): Wrong round amount, date, or investor attribution Minor (−1): Ambiguous phrasing that could be misread (e.g., "recent" for a 6-month-old round)
F4	Contact name and title are correct	Major (−2): Wrong person addressed, or wrong title used Minor (−1): Title slightly off (e.g., "Co-Founder" vs. "Co-Founder & CEO")
F5	News references are real and correctly characterized	Major (−2): Reference to a news event that does not exist in the provided data Minor (−1): Real event but slightly mischaracterized

Dimension 6: Technical Quality

Applies primarily to Ashley and Madison. For Danielle, score N/A and redistribute weight. For Tiffany, score only if the context type is technical (thesis_driven_deep_dive, cold_technical_interest).

ID	Criterion	Scoring Guide
T1	Technical terms are used correctly and specifically	5: Terms are precise and demonstrate genuine understanding (e.g., "retrieval-augmented generation" not just "AI") 3: Terms are correct but generic (e.g., "machine learning" when a more specific term applies) 1: Terms are misused, conflated, or buzzword-stuffed

T2	Thesis connection is coherent	5: Clear, specific link between the product's architecture and the investor's focus area 3: Thesis connection is present but vague 1: No thesis connection, or the connection is logically incoherent
T3	Technical depth matches available data	5: Deep-dive when data is rich; appropriately surface-level when data is sparse 3: Slightly too deep (overreaching from limited data) or too shallow (ignoring available detail) 1: Grossly mismatched depth (hallucinated architecture when data is sparse, or generic when data is rich)

Dimension 7: Actionability

Would a founder know exactly what to do after reading this email?

ID	Criterion	Scoring Guide
AC1	A clear, single ask is present	5: One unambiguous ask (meeting, call, intro) 3: Ask is present but buried or ambiguous 1: No ask, multiple competing asks, or ask is inappropriate (e.g., "are you fundraising?" on cold intro)
AC2	Ask is correctly calibrated for context	5: Cold = exploratory meeting; warm = catch-up; event = in-person meetup 3: Ask is reasonable but slightly off-caliber 1: Ask is wildly miscalibrated (e.g., requesting term sheet on cold outreach)
AC3	Ask follows demonstrated understanding	5: Ask comes after the personalization and homework section, not before 3: Ask is positioned acceptably but feels premature 1: Ask leads the email with no prior demonstration of research

5. Aggregation and Reporting

Per-Output Score Sheet

For each evaluated output, record:

- **Dimension scores** (D1–D8): integer 1–5 per dimension
- **AI-Tell violations** (D5 detail): list of specific tells detected with IDs
- **Factual violations** (D6 detail): list of specific errors with IDs
- **Weighted global score** calculated from dimension weights
- **Pass/Fail determination** based on 3.5 threshold and automatic flags

5 Automatic Flag Triggers

The following conditions trigger an automatic flag for pipeline review regardless of global score:

- Any single dimension scores 1
- D6 (Factual Accuracy) has any major violation (fabricated claims)
- D5 (AI-Tell Absence) detects 2+ violations in a single output
- Subject line is missing (F4 parsing failure)

LLM-as-a-Judge Implementation Notes

Grading Prompt Design

- **Self-contained criteria:** Each criterion (V1–V7, P1–P5, etc.) includes the full scoring definition. The judge LLM shouldn't need external context beyond the text and the email being evaluated, in which citations are included for facts.
- **Chain-of-thought:** Require the judge to output reasoning before the score (G-Eval pattern). This improves scoring reliability, especially for subjective dimensions like Voice Fidelity (from other papers on the subject).
- **Smooth scoring:** Accept the full 1–5 range per dimension. Avoid forcing pass/fail on individual criteria to preserve gradient signal for optimization.
- **Structured output:** Require JSON output with per-dimension scores, reasoning strings, detected violations (for D5 and D6), and a final weighted score.